STRUCTURAL DYNAMICS RESEARCH

Concrete Bridge Deck and Slab Fatigue under Lane-Free Traffic

Assessment Using a Vehicle–Bridge Interaction Framework

Wang Yanjun
Civil, Environmental and Resources Engineering, Faculty of Engineering, Kyoto University

Research question

How does traffic organization influence concrete deck fatigue performance under the same vehicle fleet?

Computational model

26,520

MITC4 bending/shear plate elements

Coupled steel beams and vertical supports
Approximately 79,600 algebraic coordinates
Linear equivalent model in MATLAB

Study design

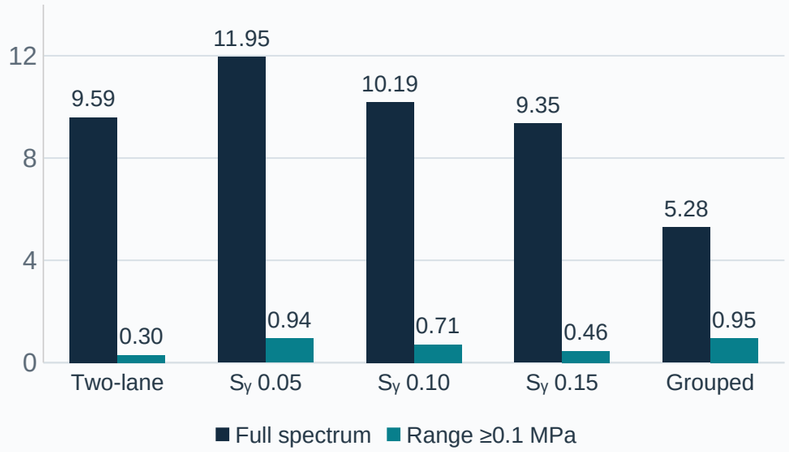
22 selected traffic scenarios use the same 240-vehicle fleet, including 20 heavy vehicles.

The comparison varies longitudinal spacing, lateral spread and vehicle ordering, with two-lane and grouped-traffic references.

Compression fatigue

Cumulative damage $D = \Sigma\, n_i/N_i$ uses the adopted concrete compression S-N relationship. The same eligible observation locations define the comparison.

Maximum damage per 240 vehicles ($\times 10^{-14}$)



d = 40 m, traffic increments, K = 17. The range cutoff tests sensitivity to small cycles. Tensile-exceedance locations remain outside the compression assessment.

3 locations / 7 episodes

Grouped traffic exceeds the tensile threshold at three common observation points.

Stress-range coverage

Nominal sections cover all saved observation coordinates.

Recorded stress ranges reach at least 98.53% of the interior field-range upper bound. This measures range-to-bound coverage.

Numerical checks

3,808 independent section-resultant checks and 23,680 cycle/parameter recomputations verify the implementation.

The maximum relative error in recalculating damage is 8.6×10^{-15}.

Fatigue interpretation

n_i: counted cycles
N_i: cycles from the adopted S-N model

Controlling damage uses S-N continuation beyond two million cycles. Small-cycle treatment affects comparisons. Tensile exceedance remains a separate outcome.

Individual work: model reconstruction, implementation, verification, simulation review and documentation. Professor Kai-Chun Chang supplied the bridge drawings.

Conditional numerical fatigue assessment. Loading, S-N continuation and cycle-cutoff assumptions limit interpretation. No actual service-life prediction.